

The Finite Part of a Hyperbolic Cone Need Not Be Closed Under Addition

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Status: candidate full counterexample.

1 The question

For an element v of a cone $\mathcal{C}$, Gigli defines its part at infinity by

$$\varepsilon v := \sup\{w \in \mathcal{C} : w \leq \lambda v \text{ for every } \lambda > 0\}, \quad \mathcal{C}_{\text{finite}} := \{v \in \mathcal{C} : \varepsilon v = 0\}.$$

Remark 3.6 of [1] asks whether $\mathcal{C}_{\text{finite}}$ is always closed under addition and says that no counterexample is known (source PDF page 38). The preceding discussion proves only $\varepsilon(v + w) \geq \varepsilon v + \varepsilon w$.

Theorem 1. *There is a cone $\mathcal{C}$ for which $\mathcal{C}_{\text{finite}}$ is not closed under addition. In fact, $\mathcal{C}$ admits a hyperbolic norm that is positive on every nonzero element and finite on every element of $\mathcal{C}_{\text{finite}}$.*

2 Construction and proof intuition

Take two copies of the open positive ray, join their origins, and put one common top above them. Addition behaves normally along either ray, but a sum involving both rays “crashes” into the top. Shrinking an element along one ray therefore leaves no nonzero common lower bound, whereas shrinking a mixed sum does nothing because the top is fixed by every positive scalar. This is exactly the strict-superadditivity mechanism that the question allows but does not exhibit.

Formally, put

$$\mathcal{C} = \{0, \infty\} \cup \{a_s : s > 0\} \cup \{b_t : t > 0\}.$$

Define $0 + x = x$, $\infty + x = \infty$, and, for positive parameters,

$$a_s + a_t = a_{s+t}, \quad b_s + b_t = b_{s+t}, \quad a_s + b_t = \infty. \quad (1)$$

For $\lambda > 0$, let

$$\lambda a_s = a_{\lambda s}, \quad \lambda b_t = b_{\lambda t}, \quad \lambda \infty = \infty, \quad (2)$$

and put $0x = 0$ for every $x \in \mathcal{C}$.

3 Verification of the cone axioms

The operations are commutative. Associativity is ordinary when all nonzero finite summands lie on one ray; as soon as both ray types occur, both parenthesizations equal ∞ . The scalar laws and both distributive laws follow from the same dichotomy. For instance, if $\lambda > 0$, then

$$\lambda(a_s + b_t) = \lambda \infty = \infty = a_{\lambda s} + b_{\lambda t};$$

the case $\lambda = 0$ gives zero on both sides. Thus $\mathcal{C}$ is a prewedge.

Its algebraic order $x \leq y \iff x + z = y$ for some z is completely described as follows:

$$\begin{aligned} 0 \leq x \leq \infty & & (x \in \mathcal{C}), \\ a_s \leq a_t \iff s \leq t, & \quad b_s \leq b_t \iff s \leq t, \\ a_s \text{ and } b_t \text{ are incomparable} & & (s, t > 0). \end{aligned}$$

It is therefore antisymmetric. A directed subset not containing ∞ cannot meet both positive rays, since their only common upper bound is ∞ . Hence it lies in one ray (apart from 0). If its parameters are bounded, its supremum is the point with the usual real supremum; if they are unbounded, its only upper bound is ∞ . A directed subset containing ∞ plainly has supremum ∞ . Thus the order is directed complete.

Finally,

$$a_s = \sup(\{0\} \cup \{a_{\eta s} : 0 < \eta < 1\}), \quad b_t = \sup(\{0\} \cup \{b_{\eta t} : 0 < \eta < 1\}), \quad \infty = \sup_{0 \leq \eta < 1} \eta \infty.$$

Any $\eta \in (0, 1)$ already gives $\eta \infty = \infty$. The wedge compatibility axiom holds, so $\mathcal{C}$ is a cone.

4 The finite part

Fix $s > 0$. If $w \leq \lambda a_s$ for every $\lambda > 0$, the order classification forces either $w = 0$ or $w = a_r$ with $r \leq \lambda s$ for every $\lambda > 0$. The latter is impossible for $r > 0$. Hence

$$\varepsilon a_s = 0, \quad \varepsilon b_t = 0 \quad (s, t > 0).$$

On the other hand, $\lambda \infty = \infty$ for every $\lambda > 0$ and every element lies below ∞ , whence $\varepsilon \infty = \infty$. Consequently

$$\mathcal{C}_{\text{finite}} = \{0\} \cup \{a_s : s > 0\} \cup \{b_t : t > 0\}, \quad a_1, b_1 \in \mathcal{C}_{\text{finite}}, \quad a_1 + b_1 = \infty \notin \mathcal{C}_{\text{finite}}.$$

This proves the negative answer.

5 A genuine hyperbolic Banach space

Define

$$\|0\| = 0, \quad \|a_s\| = s, \quad \|b_t\| = t, \quad \|\infty\| = +\infty.$$

For a nonnegative linear combination of two elements, if all positive terms lie on one finite ray, the hyperbolic norm inequality is an equality. In every other nonzero case involving both rays or ∞ , the resulting element is ∞ and the left side is $+\infty$, so

$$\|\lambda x + \eta y\| \geq \lambda \|x\| + \eta \|y\| \quad (\lambda, \eta \geq 0).$$

Thus $(\mathcal{C}, \|\cdot\|)$ is a hyperbolic Banach space under the paper's definition, with a positive norm finite on all of $\mathcal{C}_{\text{finite}}$.

6 Three neighboring obstructions

The same example sharpens the diagnosis preceding Remark 3.6:

1. The part at infinity is strictly superadditive, maximally so:

$$\varepsilon(a_1 + b_1) = \infty \quad \text{but} \quad \varepsilon a_1 + \varepsilon b_1 = 0.$$

2. The desired Riesz-style decomposition of a bound fails. Although $a_2 \leq \infty = a_1 + b_1$, there are no $z_1 \leq a_1$ and $z_2 \leq b_1$ with $a_2 = z_1 + z_2$: if both are nonzero their sum is ∞ , while if one vanishes the other cannot equal a_2 under the stated bounds.
3. Translation need not preserve infima. With $A = \{a_{1/n} : n \geq 1\}$, one has $\inf A = 0$, but

$$A + b_1 = \{\infty\}, \quad \inf(A + b_1) = \infty \neq b_1 = b_1 + \inf A.$$

In particular, the construction pinpoints why additional lattice or decomposition hypotheses can repair the theory, while the axioms of a cone alone cannot.

Novelty and scope

The latest source revision available to this run (11 December 2025) still states the question and the absence of counterexamples. Bounded searches of the run registry and of exact phrases, title, author, arXiv identifier, and citations found no later answer. Novelty is therefore provisional. The result makes no claim for the paper's stronger subclasses, such as cones with joins or cones satisfying directed decomposition.

References

- [1] N. Gigli, *Hyperbolic Banach spaces*, arXiv:2503.10467, revision of 11 December 2025.